

EDUCATION

Tulane University, School of Science and Engineering
B.S. in Mathematics, GPA: 3.97 / 4.00

New Orleans, LA
August 2024 – May 2027, Expected

RESEARCH INTERESTS

Post-quantum and code-based cryptography; algebraic coding theory; quantum information theory; algebraic geometry in cryptography.

PUBLICATIONS & PREPRINTS

- [1] M. B. Can and E. C. Naig, “A Theory of Leakage in Secret Sharing and Evaluation Codes”, Submitted, under review, 2026.
- [2] M. B. Can and E. C. Naig, *Applications of Bruhat-Chevalley-Renner Decomposition to Metric-Aware Code-Based Cryptography*, Cryptology ePrint Archive, Paper 2026/665, Preprint, 2026. [Online]. Available: <https://eprint.iacr.org/2026/665>
- [3] M. B. Can and E. C. Naig, “Module-HQC”, Submitted to IEEE Transactions on Information Theory, 2026.
- [4] E. C. Naig and S. A. Fenner, *Parallel Repetition in the Two-Player Quantum Cloning Game*, arXiv:2608.00930 [quant-ph], Preprint, 2026. [Online]. Available: <https://arxiv.org/abs/2608.00930>

TALKS & PRESENTATIONS

- *Quantum Position Verification and the Quantum Cloning Game*. Poster, USC Summer Research Symposium, University of South Carolina. July 2026
- *Quantum Position Verification: Cloning Games and an Open Repetition Problem*. Talk to faculty and students of the NSF REU in Quantum Information Science, University of South Carolina. July 2026

PATENTS

- **Systems and Methods for Secure and Reliable Data Storage and Transmission** Filed April 2026
PCT patent application; co-inventors: Mahir Bilen Can, Dillon Montero, and Eli Coe Naig. Tulane Ref. No. 2025-010-02; Foley Ref. No. 136669-0184.
- **Metric-Aware Code-Based Cryptography** Filed March 6, 2026
U.S. provisional patent application; co-inventors: Mahir Bilen Can and Eli Coe Naig. Attorney docket no. 752001-8010.

RESEARCH EXPERIENCE

Undergraduate Researcher

Advisor: Mahir Bilen Can, Tulane University

New Orleans, LA

August 2025 – Present

- Designed code-based public-key cryptosystems, including a quasi-circulant trapdoor for Niederreiter-type key encapsulation.
- Extended code-based cryptography beyond the Hamming metric to the NRT and sum-rank metrics, using code isometry groups and Bruhat-Chevalley-Renner decompositions.
- Proved an NP-hardness theorem and gave decoding-hardness reductions underpinning the schemes' security.
- Developed a matroid-based theory of information leakage in linear secret sharing and evaluation codes.

NSF REU in Quantum Information Science

Advisor: Stephen Fenner, University of South Carolina

Columbia, SC

May 2026 – August 2026

- Studied quantum position verification at USC's NSF REU site (NSF award 2447817).
- Disproved strong parallel repetition for the quantum cloning game and improved its best known upper bound (see Publications).

RTG Summer Math Program

Louisiana State University

Baton Rouge, LA

May 2025 – June 2025

- Studied hyperplane arrangements and matroid theory in the workshop funded by LSU's NSF Research Training Group grant (NSF award DMS-2231492).
- Prepared and delivered lectures on hyperplane arrangements, geometric lattices, and characteristic polynomials.
- Led peer discussions linking combinatorial geometry to coding theory.

TECHNICAL & ENTREPRENEURIAL EXPERIENCE

QSPARC Labs

Co-Founder, CEO, and Lead Developer

New Orleans, LA

August 2025 – Present

- Co-founded a startup developing post-quantum cryptography grounded in original coding-theory research.
- Build reference implementations of the published schemes in Python, Rust, and C, with test suites and reproducible benchmarks.
- Verified a post-quantum key-establishment protocol with machine-checked symbolic security proofs in ProVerif.
- Direct the company's patent work: two applications filed, with more in preparation (see Patents).
- Lead and mentor a team of four: two engineering interns and two part-time developers.
- Accepted into an NSF-funded I-Corps-style commercialization program administered through MIT.

TEACHING & MENTORING

- Volunteer Tutor, Tulane University

Fall 2025 – Present

Tutor student-athletes in linear algebra and abstract algebra, supporting coursework alongside practice and competition schedules.

AWARDS & GRANTS

- Tulane Honors Scholar 2024–2028
- Dean’s List, Tulane University Fall 2024; Spring 2025; Fall 2025; Spring 2026
- CELT Faculty-Mentored Undergraduate Research Grant, Tulane University; mentor: Mahir Bilen Can. 2026
- Offered a place in the Newcomb-Tulane College Summer Research Program; declined in favor of the NSF REU. 2026
- Chosen for a Tulane Innovation Institute program on research translation and proposal development. 2025

LEADERSHIP & SERVICE

- President, Tulane Math Club 2026–2027
Lead programming and organization for the university math club.
- Co-Founder and Vice President, Tulane Science Olympiad 2024–Present
Co-founded Tulane’s first Science Olympiad invitational; coordinated logistics for an event serving 380 students from nearly 20 schools and developed tulanescioly.org.
- Trip Leader, Tulane Outdoor Adventures 2024–Present
Lead outdoor trips for student groups, managing route planning, risk assessment, logistics, and participant safety.

TECHNICAL SKILLS

- **Programming:** Python, Rust, C, C++, Java, JavaScript, TypeScript, MATLAB, R
- **Tools:** Git, GitHub, L^AT_EX, Linux, ProVerif
- **Languages:** English (native), Chinese (intermediate reading/writing; beginner speaking)

SELECTED ADVANCED COURSEWORK & INDEPENDENT STUDY

- **Selected Advanced Coursework:** Abstract Algebra I–II, Real Analysis I–II, Linear Algebra, Probability, and Combinatorics.
- **Planned Fall 2026 Coursework:** Algebra I, Algebraic Geometry, Mathematical Foundations of Computer Security, and Honors Thesis.
- **Independent Study:** Algebraic geometry, error-correcting codes, real analysis, and manifolds; texts include Hartshorne, Huffman–Pless, Rudin, and Spivak.